A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Sampling techniques & measuring abiotic factors - Set 1

Mark scheme

AQA spec 3.7.2.1-3.7.2.2; Research methods; textbook pp. 394-419

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora **1** 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants) – 100 m x 100 m is for trees (impractical, too much work) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = 240 – N(N-1) = 240 x 239 = 57 360 **1** Σn(n-1) = 48x47 + 120x119 + 36x35 + 24x23 + 12x11 = 2256 + 14280 + 1260 + 552 + 132 = 18 480 **1** D = 57 360 / 18 480 = 3.10 (accept 3.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance • **Biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1 Lower D means lower biodiversity:** the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) **1** Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **plagioclimax** → a community held back from its climax by human activity	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants **1** Point quadrat pins are lowered through the vegetation – each touch counts (10 pins = 10% each) without flattening **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
01.5	1 mark for the point below. **1** Subjective judgement varies between recorders – results are not quantitative so statistical analysis is not possible (but quick for many areas) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	

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Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Net held with its straight edge on the river bed facing the current (open to the flow) **1** The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) **1** Same substrate type, water depth and flow rate at both sites – same net mesh – repeat samples – invertebrates identified with a key and counted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	3
02.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Difficult to standardise the vigour / area kicked **1** Buried organisms or those fixed to stones are not dislodged **1** Some organisms swim and avoid the net **1** Flow rate varies and affects capture **1** Fragile organisms destroyed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed area of bed is enclosed by a frame so the sampled area is standardised – side nets reduce escapes **1** But only works in fast enough flow and is hard to set on rough substrates / deep water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	2

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Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event **1** Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Ground surface:** pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals – trapped predators eat others – preservative attracts / repels – escapes **2 Leaf litter:** Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light – some die or stop moving – large ones cannot pass the mesh **2 Branches:** beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour – some species not dislodged / fly away – high branches unreachable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pitfall trap** → a sunken container that catches ground invertebrates • **Tullgren** → a funnel that drives soil invertebrates down into a collecting jar	6
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught **1** Brightness / wavelength of the lamp affects attractiveness so catches vary between traps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** The area of ground covered by the suction pipe is known so invertebrates per m^2 can be calculated **1** A pooter is only a handling / collection device for individuals already found, not a sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
04.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (60 x 45) / 9 **1** = 300 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (beetles) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
04.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Marked individuals mix freely / randomly with the rest of the population **1** No births, deaths, immigration or emigration between samples **1** Marking does not affect survival or behaviour (eg make them more visible to predators) **1** Marks do not wear off / fall off **1** All individuals have the same chance of being caught (traps do not attract or repel marked animals) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Lincoln Index** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	4

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Fewer marked beetles would survive to be recaptured **1** So the proportion marked in the second sample would be too low and the population would be overestimated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Enough time for marked beetles to mix with the population **1** Not so long that births / deaths / migration or loss of marks change the population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** Readings drift so they may not be accurate – calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions **1** pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly **1** If a meter cannot be adjusted, record the percentage error and correct the results **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	3
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Same height above the ground **1** No obstacles to air flow – impeller type facing into the wind **1** Axis of rotation horizontal (impeller) or vertical (cup) – same time period / averaging **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling **1** Evaporation cools the wet bulb more when humidity is low – the temperature difference is compared with a table to find humidity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
05.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Test strips with reagent bands compared with a colour chart (semi-quantitative) **1 Electronic colourimeter:** reagent added and colour intensity measured at a specific wavelength **1** Ion-selective electrode measuring electrode potential (other ions can interfere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Sieves assembled with the coarsest mesh on top – a dried, crushed soil sample is placed in the top sieve **1** Stack shaken (about 2 minutes) until no more particles pass – each fraction weighed **1** Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated – material over 2 mm (stones, twigs) is not soil – plotted on a soil triangle **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Auger collects a core of known cross-sectional area and depth **1** So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals such as calcium carbonate break down releasing carbon dioxide **1** So mass is lost that is not organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** A beam of light passes through the sample in a dark chamber – a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter **1** Standardise bulb brightness / colour, distance and container **1** A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations – Secchi discs / turbidity bottles give only semi-quantitative results **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable **1** Wild species have different ranges of tolerance so cress results cannot be applied directly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **tolerance** → the range of conditions in which a species can survive **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Preliminary study with widely spaced pH values to find where germination occurs **1** Final range from below the lowest pH giving germination to above the highest, with closer intervals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 4 points below. **1** Temperature (constant, natural germination temperature) **1** Light, water volume, number of seeds per dish, seed source **1** Growth medium (filter paper; soil would alter pH) – nutrients not needed for germination **1** Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

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Question	Marking guidance	Marks
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface water affects fresh mass and drying damages the plants – dry mass requires killing the plants so the same plants cannot be re-measured **1** Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1 Choosing by eye is biased:** people are drawn to patches that look interesting, flowery or easy to reach, so the sample does not represent the whole field **1** Random numbers used as coordinates on a grid give every point an equal chance of being sampled, so the sample is representative and the mean can be applied to the whole area **1** An unbiased sample makes the comparison between the two fields valid, because any difference found is due to the grazing and not to how the sites were chosen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area • **random** → positions chosen using random numbers so the choice is not biased • **valid** → actually measures what it is meant to measure **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	3
08.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = (number in first sample x number in second sample) / number marked in second sample = (60 x 75) / 15 **1** = 300 grasshoppers **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Lincoln index** → estimating population size from the proportion of marked animals recaptured **Do not credit** • A bare wrong answer with no working.	2

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Question	Marking guidance	Marks
08.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The marked individuals mix randomly back into the population before the second sample is taken **1** The mark does not affect the animal's survival or behaviour, is not lost, and does not make it more visible to predators or to the collector **1 The population is closed:** there are no births, deaths, immigration or emigration between the two samples **1** Marked and unmarked individuals are equally likely to be caught in the second sample **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **valid** → actually measures what it is meant to measure **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	3
08.4	**1 mark each** 1 mark for each of the 3 points below. **1** The quadrat must be large enough to contain a workable number of individuals of the species present but small enough to count accurately - a small quadrat for low-growing grassland plants, a larger one for shrubs **1 A pilot study is used:** the cumulative number of species found is plotted against the number of quadrats, and enough quadrats are taken for the curve to level off **1** More quadrats reduce the effect of chance on the mean and make the result more reliable, but the time available limits how many can be done, so a compromise is reached **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3

Do not write outside the box

Question	Marking guidance	Marks
08.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Survey both fields on the same day, or in the same weather, so that differences in temperature, rainfall or season do not affect the result **1** Use the same method, the same quadrat size and the same number of samples in each field, so the only difference between them is the grazing **1** Choose fields that match in soil type, aspect, slope and altitude, so grazing really is the only variable being tested **1** Identify the plants with the same key, and have the same person make the estimates, so judgements such as percentage cover are consistent **1** Repeat the whole survey and calculate means, and record the spread of the data so the variability within each field can be compared with the difference between them **1** Avoid trampled areas near gates and the field edges, where conditions are untypical of the field as a whole **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **reliable** → repeatable / gives consistent results when repeated • **valid** → actually measures what it is meant to measure • **quadrat** → a square frame of known area used to sample a fixed area **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-402; AQA spec 3.7.2.1-3.7.2.2.	

Do not write outside the box